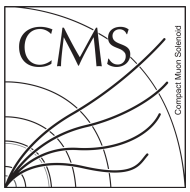
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The Compact Muon Solenoid Experiment
CMS Performance Note



MCP-referenced timing calibration of CMS BTL

CMS

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Abstract

Several analyses of the 2025 data set benefit from an external time reference (t_0). During the testbeam period a Hamamatsu MCP-PMT was placed downstream of the setup and waveforms were digitized on a Lecroy scope. Since the active area ($\emptyset$=10mm) of the PMT only covered a couple BTL bars under test and the scope DAQ captured only 7.5k events per spill, we used a limited data set to calibrate Reference Bar timing relative to the MCP (which had intrinsic resolution of 7.5 picoseconds). The calibration functions are provided in this note.

Draft v1 for scientific discussion. Not approved for public release.
The following slides retain explicitly marked review items.